

DOCUMENT 04

Detection and model integration

CTD + COO, multilingual coverage and optional segmentation experiments

1. Recommended experiment, not a guaranteed winner

Working direction: CTD + a COO-trained DBNet++ detector, with no recognition stage. Retain ogkalu for its existing layout/candidate role. Feed all accepted candidates into the text-shaped mask workflow in Document 03.

The model components have different responsibilities:

Component	Role in the proposed system	Status
CTD	Baseline text discovery and mask evidence	Retain
ogkalu text/bubble detector	Enclosure and complementary text candidates	Retain
COO-trained DBNet++	Additional irregular SFX discovery	Benchmark first
Refined CTD mask	Initial lettering-mask baseline	Implement and evaluate
SAM-TS / Hi-SAM stroke output	Difficult-mask comparator	Optional experiment
PP-OCRv6 medium detector	Additional non-recognition text discovery	Reserve experiment
manga-ocr / TRBA	Reading text	Not required for this workflow

What COO's published evidence establishes

COO targets Japanese comic onomatopoeia and supplies text-region polygons and fragment links. The repository lists detection checkpoints and separate recognition/linking tasks. [\[S22\]](#)

COO checkpoint	Precision	Recall	Hmean
DBNet++	90.8%	60.9%	72.9%
MTSv3, detection-only evaluation	70.1%	66.0%	68.0%
ABCNetv2, detection-only evaluation	67.2%	65.1%	66.1%

These are repository-reported single-checkpoint results, not app measurements. DBNet++ has the highest Hmean among these rows, but not the highest recall. Different thresholds and combinations may change the relevant tradeoff. [\[S22\]](#)

No inference from these numbers establishes performance on Korean lettering. The user's color-webtoon example requires a separate Korean holdout. Do not ship a Japanese-only success claim as an “everything” result.

2. Candidate generation and fusion

Discovery passes

B1 - Preserve a full-page pass. It can see effects or layout extending beyond any one crop.

B2 - Add overlapping crops as a quality profile. Test the earlier 1024-1536 source-pixel, 15-25% overlap proposal against actual source dimensions and device budgets. Do not squeeze an entire very long webtoon into a small square and call that the only high-recall pass.

B3 - Evaluate unused CTD evidence. Implement line-map decoding and a conservative path for orphan segmentation components. Upstream demonstrates these mechanisms; their benefit in this app still needs measurement. [S20]

B4 - Run complementary discovery beyond existing boxes. A detector applied only within CTD-found regions cannot discover wholly missed text elsewhere on the page.

B5 - Map once to native coordinates. Store crop origin, letterbox/resize transform and source/decode-window identity. Reconcile overlap duplicates in page coordinates.

Fusion policy

Preserve credible candidates from either detector. Do not require ensemble agreement. Use per-model calibrated thresholds; raw scores from different architectures are not directly comparable.

Resolve duplicates using overlap plus text evidence and enclosure relationships. Do not suppress a text candidate because it lies inside a bubble candidate: the container and its text are different objects. Avoid merging two adjacent lines or unrelated effects solely because padded boxes intersect.

Preserve candidate lineage after merging, including model/checkpoint, raw score, crop and polygon. Store SFX as a proposed role only when justified; a generic “outside bubble” label does not establish sound-effect semantics.

Fragment handling without OCR

A visual effect can be one selectable group with several masks. Use spatial/appearance evidence or manual grouping; no transcription is required. Keep separate components across large gaps. Recognition and learned link prediction are not required to clean each fragment.

Failure outcome: a region discovered only by COO remains visible even when segmentation fails. Mark it “mask needed,” rather than dropping the detection or erasing its polygon automatically.

3. Segmentation candidates and their limits

DB polygons do not solve the mask requirement

DB's postprocessing converts a probability map into boxes/polygons and expands detected text regions. These are localization outputs, not a guarantee of exact glyph pixels. [S24]

Do not rasterize the entire COO polygon as the default new-mode erase area. Use it to select a crop, run a stroke segmenter and associate the resulting mask with the candidate. A rectangular crop is acceptable; a rectangular write mask is the behavior being avoided.

Comparator A: refined CTD

Start here because the app already has mask evidence. Measure source-image refinement and local segmentation reruns separately from detection additions. Keep CTD mask threshold, source refinement and user padding independently recorded.

Comparator B: SAM-TS / Hi-SAM

Use the **stroke-level** output, not a word, line or paragraph envelope. The official project provides automatic text-stroke inference and a sliding-window option. The demonstration distinguishes stroke segmentation from hierarchical detection. [S27, S36]

Choose a base-sized checkpoint for the first cost/quality experiment; consider the efficient variant only after output semantics and quality are checked. There is no verified out-of-box Korean-comic superiority, no app latency measurement, and no confirmed native export in this report.

Comparator C: ogkalu comic segmenter

Inspect mask granularity and actual coverage on outlined Korean and Japanese SFX. A model called "segmenter" may still use broader region annotations. The short card alone is insufficient evidence for replacing the current mask source. [S26]

Reserve discovery model: PP-OCRv6

The official medium detector has an ONNX release. A detector can be used without its recognizer. It remains an optional coverage comparison, not a required new stage or a promise of manga-specific SFX accuracy. [S30]

Promote only on incremental benefit

A model earns a production role when it adds meaningful recoveries or reduces correction time without unacceptable artwork exposure, latency or memory. Do not automatically load every researched model. Detailed semantic classification and VLM review remain deferred.

4. Prototype, export and deployment gates

Research prototype

Use an isolated Python environment for checkpoint evaluation and save plain candidate/mask outputs. Do not integrate a research stack into the desktop installer merely to prove the idea. Pin code, configuration and checkpoint hashes for reproducibility.

The DB implementation and COO's MTSv3/ABCNetv2 setups include research dependencies and compiled operations. [S23, S32, S33] Export success, supported operators and provider behavior must be established for the exact checkpoint, not inferred from a model family name.

Native adapter contract

The production adapter should return candidates with geometry, confidence, model identity and source transforms. A separate segmenter adapter should return per-pixel evidence with explicit semantics. Model input color order, normalization, padding, tensor names/shapes, output thresholds and coordinate conventions belong in the artifact manifest.

Validate Python-versus-export parity on fixed inputs before quality comparisons. Include CPU and the actual execution providers intended for each supported desktop platform. Compare outputs within declared numerical tolerances and inspect any changed candidate/mask decisions.

Optional model management

Use the app's model-manager pattern for download permission, checksums, on-demand residency, missing-model messaging and cancellation. A missing optional SFX detector must leave the baseline available. A no-recognition run must not be blocked by missing OCR or script-gate files.

Keep checkpoints and source-image rights separate. The COO DBNet++ checkpoint's redistribution status must be established explicitly; a code license or publicly downloadable weight is not a blanket clearance.

Required benchmark slices

Japanese monochrome; Korean color webtoon; ordinary bubble text; outlined/rotated/curved SFX; white and low-contrast text; tiny text/furigana; punctuation-only text; text crossing artwork; long-strip joins; mixed-script pages. Use unseen series where feasible.

Ship/no-ship rule: a specialist should materially improve the target corpus over the refined baseline. If COO fails the Korean slice, retain the plugin boundary and improve the dataset/model rather than asserting language independence.

Source IDs link directly to primary references. Full evidence ledger: Document 08.